Fiscal Year:	FY 2020	**Task Last Updated:**	FY 09/15/2020
PI Name:	Nicholson, Wayne Ph.D.		
Project Title:	Global Transcriptome Profiling to Identify Cellular Stress Mechanisms Responsible for Spaceflight-Induced Antibiotic Resistance		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology		
Space Biology Cross-Element Discipline:	(1) Reproductive Biology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
PI Email:	WLN@ufl.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	321-261-3773
Organization Name:	University of Florida		
PI Address 1:	Space Life Sciences Laboratory		
PI Address 2:	505 Odyssey Way, Room 201-B		
PI Web Page:			
City:	Merritt Island	**State:**	FL
Zip Code:	32953	**Congressional District:**	3
Comments:	NOTE: Congressional District is for University of Florida (32611-0001)--Ed., 8/17/2012; changed to 3 (Ed., 7/30/13)		
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2014 Space Biology Flight NNH14ZTT001N
Start Date:	10/01/2014	**End Date:**	09/30/2021
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	3	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA KSC
Contact Monitor:	Levine, Howard	**Contact Phone:**	321-861-3502
Contact Email:	howard.g.levine@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	ISS NOTE: End date changed to 9/30/2021 per NSSC information (Ed., 9/10/20) NOTE: Extended to 9/30/2020 per NSSC information (Ed., 10/24/2019) NOTE: Extended to 9/30/2019 per NSSC information (Ed., 1/11/2019) NOTE: Extended to 9/30/2018 per NSSC information (Ed., 10/11/17)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Fajardo-Cavazos, Patricia Ph.D. (University of Florida)		
Grant/Contract No.:	NNX14AT38G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Responses of various microorganisms to the spaceflight environment have been examined by numerous studies conducted either in space or on the ground in clinostats. The results clearly show that several microbial behaviors differ in space, but not in a consistent way. The underlying mechanism(s) involved are not explainable by a single model. Exposure to the spaceflight environment (e.g., microgravity, radiation) alters a microbial cell's immediate surroundings, lowering convective mass and heat transfer or reducing mechanical shear forces. Changes in such fundamental physical forces affect the rates at which gases, nutrients, signaling molecules, and waste products are exchanged between microbes and their surroundings. Microbes perceive these alterations as environmental stress (i.e., the "spaceflight syndrome") and mount a complex set of stress responses. One consequence of the spaceflight stress response is alteration in their susceptibility to antibiotics. This could be due to mechanisms such as: production of biofilms, alteration of cell surfaces, up-regulation of drug efflux systems, or an increased rate of mutation in genes encoding drug targets. All of these mechanisms have been well documented as outputs of global microbial stress responses. Of particular interest is oxidative stress, which has been shown both to trigger maladaptive responses in higher organisms including astronauts, and to increase bacterial resistance to antibiotics. Central Hypothesis: Microorganisms subjected to the integrated spaceflight environment invoke a spectrum of stress responses, some leading to alterations in their antibiotic susceptibility. The underlying mechanisms can be identified using transcriptome profiling of model organisms. Specific Aims: We postulate that spaceflight stress responses vary depending on the organism studied because each species of microbe is endowed with its own unique collection of stress response systems. In order to probe the underlying mechanism(s) involved it will be necessary to study spaceflight stress response at the genomic level using model organisms. Specifically we plan to: A. Expose the model bacteria Bacillus subtilis and Staphylococcus epidermidis to: (i) simulated microgravity in the laboratory and (ii) actual spaceflight in 2 parallel Biological Research in Canisters (BRIC) canisters (BRIC-A and BRIC-B) aboard the International Space Station (ISS), with appropriate ground controls. Cells cultivated in BRIC-A will be fixed rapidly with RNA Later and cells in BRIC-B will be frozen in a viable state for return to Earth. B. Total RNA will be extracted from cells in BRIC-A and subjected to transcriptome analysis using Whole Transcriptome Shotgun Sequencing (a.k.a. RNA sequencing or "RNA-Seq") at the University of Florida Interdisciplinary Center for Biotechnology Research (UF-ICBR), to identify the suite of stress responses induced by exposure to spaceflight or simulated microgravity. Cells in BRIC-B will be thawed and immediately assayed for (i) viability, (ii) resistance levels to a battery of antibiotics in the Omnilog system, and (iii) rates of spontaneous mutation to antibiotic resistance.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Exposure of microorganisms to spaceflight provokes a variety of stress responses, some of which have been postulated to lead to increased antibiotic resistance. The central goal of this proposal is to identify these stress responses, using high-throughput global transcriptome and phenotypic analyses. Understanding how potentially pathogenic microbes might develop resistance to antibiotics in response to stress has significance both for long-term human space exploration and for medical science on Earth.
Task Progress:	Introduction: This report documents the activities related to “Global transcriptome profiling to identify cellular stress mechanisms responsible for spaceflight-induced antibiotic resistance.” The original grant was to perform the spaceflight experiment “Biological Research in Canisters #21 (BRIC-21)," with the aim of understanding how bacterial cells perceive and respond to the human spaceflight environment. We have successfully completed all of the original aims and goals of this grant proposal. New Objective: In 2019, this grant was tasked with a new objective: to develop and test in spaceflight a new sample compartment for use in Biological Research in Canisters-Petri Dish Fixation Unit (BRIC/PDFU) and/or BRIC-LED (Light Emitting Diode) hardware. In all previous BRIC experiments, the sample compartment has consisted of the bottom half of a standard 60-mm Petri dish. However, during the BRIC-21 experiment it was noted that liquid cultures assumed a toroidal shape in flight (FL), in contrast to the disk-shaped ground control (GC) samples. It was hypothesized that this difference in culture geometry might account in part for the measured differences in the transcriptomic responses between FL and GC samples (Morrison, M.D., P. Fajardo-Cavazos, and W.L. Nicholson. 2019. Comparison of Bacillus subtilis transcriptome profiles from two separate missions to the International Space Station. npj Microgravity 5:1. doi: 10.1038/s41526-018-0061-0). In collaboration with Drs. Mark Weislogel and Titel Jurca, we have conceived and built a new sample compartment designed to mitigate the effects of microgravity on liquid culture geometry in petri dish fixation units (PDFUs). Drop-tower and biocompatibility testing of a prototype device was in progress when our labs were closed due to the COVID-19 pandemic, which essentially halted all progress on the project for several months.
Bibliography Type:	Description: (Last Updated: 07/25/2025)
Abstracts for Journals and Proceedings	Leehan JD, Nicholson WL. "Mutations in the Bacillus subtilis rpoB gene exhibit varying degrees of fitness in minimal and rich media." 2020 Meeting of the Society for Molecular Biology and Evolution (SMBE 2020), Quebec City, Canada, Meeting canceled due to COVID-19 pandemic. 2020 Meeting of the Society for Molecular Biology and Evolution (SMBE 2020), Meeting canceled due to COVID-19 pandemic. , Jul-2020
Articles in Peer-reviewed Journals	Morrison MD, Fajardo-Cavazos P, Nicholson WL. "Comparison of Bacillus subtilis transcriptome profiles from two separate missions to the International Space Station." npj Microgravity 2019 Jan;5:1. eCollection 2019. https://doi.org/10.1038/s41526-018-0061-0 ; PubMed PMID: 30623021; PubMed Central PMCID: PMC6323116 , Jan-2019

Articles in Peer-reviewed Journals	Morrison MD, Nicholson WL. "Comparisons of transcriptome profiles from Bacillus subtilis cells grown in space vs. High Aspect Ratio Vessel (HARV) clinostats reveal a low degree of concordance." Astrobiology 2020 Dec 14;20(12):1498-509. Epub 2020 Oct 19. https://doi.org/10.1089/ast.2020.2235 ; PMID: 33074712 , Dec-2020